

The Artist, the Track and the Lyrics: Predicting Music Success

Ivan La Franca*

Fall 2024

Abstract

Predicting the success of music tracks is a multifaceted challenge influenced by factors like artist popularity, production quality, and audience engagement. This study explores whether lyrics can serve as reliable predictors of a song's success. Leveraging data from the Genius API and Kaggle, I applied topic modeling, sentiment analysis, and an array of rhetorical and stylistic metrics to evaluate their impact on stream counts. Despite employing various predictive models, including Neural Networks and Random Forests, the results showed limited success, underscoring the need for larger datasets and more comprehensive predictors. This research highlights the complexities of quantifying music success and provides a foundation for future studies to expand upon, potentially enabling artists and record labels to optimize track selection and promotion strategies.

*IDEA MRes Student: Universitat Autònoma de Barcelona and Institut d'Anàlisi Econòmica.

1 Introduction

The music industry has experienced a seismic shift in the digital era, primarily due to the rise of streaming platforms like Spotify, Apple Music, and YouTube. These platforms have not only transformed how audiences consume music but have also generated vast amounts of data on listening behaviors, preferences, and trends. This abundance of data presents a unique opportunity to apply machine learning techniques to understand and predict various aspects of the music industry, including the success of individual tracks.

Predicting the success of a song is a complex task that holds significant economic implications for artists, record labels, and streaming services. Accurate predictions can inform marketing strategies, optimize promotional efforts, and guide investment decisions. Traditionally, factors such as artist reputation, production quality, and marketing budgets have been considered primary drivers of a song's success. However, the intrinsic content of the music, particularly the lyrical component, remains an underexplored area in predictive modeling.

This study investigates whether the lyrical content of songs, analyzed through advanced natural language processing (NLP) techniques, can serve as a reliable predictor of music success. By examining themes, sentiments, and rhetorical devices within lyrics, I aim to uncover patterns that correlate with a song's popularity as measured by streaming counts. The central question I address is whether song lyrics can reveal insights into a song's potential success.

My motivation stems from the intersection of machine learning and economics. From a machine learning perspective, songs provide rich, unstructured textual data suitable for NLP applications. Economically, understanding the impact of lyrical content on a song's success can influence how resources are allocated in the music industry. If certain themes or stylistic elements are associated with higher success rates, artists and producers can tailor their work accordingly to meet audience preferences.

The study contributes to the literature by combining NLP techniques with predictive modeling in the context of the music industry—a relatively novel approach. I employ methods such as Latent Dirichlet Allocation (LDA) for topic modeling and transformer-based models like DistilBERT for sentiment analysis. Additionally, I analyze rhetorical figures and stylistic elements, including alliteration, anaphora, and rhyme schemes, to assess their potential influence on a song's popularity.

The structure of the paper is as follows. In Section 2, I provide a comprehensive motivation and literature review, situating my research within existing studies in machine learning and economics. Section 3 details the data sources and feature engineering processes, including how I collected and preprocessed the lyrical and streaming data. In the same section, I present my text analysis findings, highlighting the thematic and sentimental attributes of the songs examined. Section 4 discusses the predictive models used, such as Neural Networks and Random Forest models, and evaluates their effectiveness in forecasting music success based on lyrical content. Finally, Section 5 offers conclusions, discusses the limitations of my study, and suggests avenues for future research.

By exploring the predictive power of song lyrics, this research aims to provide valuable insights for stakeholders in the music industry and contribute to the academic discourse on the applications of machine learning in economics.

2 Motivation and Literature Review

2.1 Motivation

Predicting the success of music tracks is a high-stakes endeavor with significant implications for both the music industry and the broader field of applied machine learning. From an economic perspective, optimizing song selection and promotion strategies can yield substantial returns for record labels, streaming platforms, and artists. Advances in machine learning and the proliferation of streaming data have opened new avenues to explore predictive modeling in this domain. However, much of the existing research focuses on acoustic features and market dynamics, leaving the potential of lyrical content underexplored.

This study addresses two key gaps. First, it provides an analytical framework that leverages natural language processing (NLP) tools to quantify the thematic, emotional, and stylistic dimensions of song lyrics. Second, it explores the value of these features by linking them to track performance metrics, such as streaming counts. By doing so, the research aims to contribute to the growing literature at the intersection of computational musicology, machine learning, and cultural economics.

2.2 Literature Review

The ability to predict cultural and economic success in creative industries, such as music, has been a longstanding research challenge. Advances in machine learning, the digitization of markets, and the availability of large datasets have enabled more precise modeling of factors influencing success. Below, key contributions are discussed. S. Rosen (1981) provides a theoretical basis for understanding how small differences in quality can lead to disproportionately large differences in market outcomes. This phenomenon is highly relevant to music, where talent, visibility, and market reach amplify success disproportionately, explaining why a few artists dominate streams and sales. Building on Rosen’s insights, Brynjolfsson, Hu, and Smith (2006) analyze the impact of digitization on consumer behavior and market structure. They demonstrate how digital platforms reduce production and distribution costs, allowing niche content to coexist with mainstream hits. This long tail effect highlights the potential for underrepresented lyrical or stylistic themes to drive music success in the streaming era. The digitization of music has also reshaped how consumers discover and engage with music. Aguiar and Waldfogel (2018) expand this perspective, illustrating that digitization not only increases the variety of available music but also enhances consumer welfare through access to previously unpredictable successes, linking product characteristics, such as lyrics, to realized outcomes. Rosen’s and Aguiar and Waldfogel’s frameworks are complemented by the digitization-driven models discussed by Brynjolfsson et al., which highlight the interplay between consumer choice and product characteristics. These insights are particularly valuable when combined with advanced NLP tools like LDA, which enable a deeper understanding of how lyrical content influences success.

Advances in machine learning have significantly enhanced the ability to predict success in creative industries, including music. For instance, Fell and Sporleder (2014) demonstrates the potential of NLP techniques such as topic modeling and sentiment analysis in understanding listener engagement with lyrics. Similarly, Malmi et al. (2016) explores how stylistic features, such as rhymes and repetitions, can inform audience reception. These contributions highlight the efficacy of machine learning tools in modeling the multifaceted determinants of music success, providing a methodological foundation for this study.

3 Data and Feature Engineering

This study relies on two primary data sources to investigate the potential predictors of music success: a dataset from Kaggle and a collection of song lyrics obtained via the Genius API. Together, these datasets provide a foundation for analyzing various textual, linguistic, and quantitative features. However, the scope and limitations of the data should be considered carefully, as they influence the generalizability of the results. Below, I describe each dataset in detail and deal with the feature engineering, followed by a discussion of the merged dataset.

3.1 Genius Data

The first dataset comprises the lyrics of 2,100 songs collected using the Genius API. Genius is a comprehensive platform for song lyrics, and this dataset represents 50 of the most-read lyrics from each of the top 42 global artists, ordered by their popularity on the platform. While the dataset provides valuable insights into the language and themes of popular music, it is inherently limited in scope. It includes only songs from the most popular artists and lacks a representation of less successful songs, making it challenging to generalize findings beyond this elite sample.

To make the lyrics suitable for analysis, I performed several preprocessing steps. Since most songs are in English, I identified and translated the lyrics of the seven Spanish-speaking artists using Google Translator’s Python library. This allowed me to standardize the data for further analysis. Table 1 provides a sample comparison between translated and original lyrics.

Translated Lyrics	Original Lyrics
<i>The successful factory</i>	<i>La fábrica de éxitos</i>
<i>The successful factory</i>	<i>La fábrica de éxitos</i>
<i>I will make the story to Pitbull and sensible to know it</i>	<i>Le voy a hacer la historia a Pitbull y a Sensato pa’ que la conozcan</i>
<i>Yuli the table is served</i>	<i>Yuli la mesa está servida</i>
<i>The soup is hot</i>	<i>La sopa está caliente</i>

Table 1: Comparison of Translated and Original Lyrics. Pitbull, El Taxi.

After translating the lyrics, I performed a topic analysis using LDA. Introduced by Blei, Ng, and Jordan (2003), LDA is a machine learning and natural language processing model designed to uncover hidden topics within a collection of text documents—song lyrics in this case. As an unsupervised learning algorithm, it identifies patterns without relying on predefined labels or output variables. LDA operates on the assumption that documents are mixtures of topics and that each topic is a probabilistic distribution of words, inferring these distributions directly from the data.

In this study, each song lyric is treated as a document, and LDA identifies the underlying topics across all songs. Each word in a lyric is assigned a probability of belonging to a specific topic, with the topic itself represented as a distribution over words. For instance, an emotionally driven topic might include words such as "love," "heart," and "feel," while a celebration-focused topic might include "party," "dance," and "fun."

By leveraging probabilistic distributions, LDA determines the likelihood of words within a topic and the proportion of each topic within a document. This enables each lyric to be described as a blend of multiple topics (e.g., 60% Party, 40% Street Life), while each topic is defined by a set of key words and their associated probabilities. LDA is particularly

well-suited for analyzing song lyrics, as it effectively uncovers thematic patterns across texts and highlights differences between translated and original content.

Finally, I obtain five topics: Love, Party, Desire, Reflection, and Street-Life. Table 2 shows number of artist by dominant topic and the first 5 words per topic.

Dominant Topic	N. of Artists	Top 5 Words
Desire	3	<i>want, feel, baby, without, life</i>
Love	11	<i>heart, fire, eyes, life, feel</i>
Party	5	<i>baby, want, tonight, crazy, dance</i>
Reflection	12	<i>think, baby, feel, find, remember</i>
Street Life	11	<i>bitch, fuck, shit, nigga, baby</i>

Table 2: Summary of Dominant Topics with Artist Counts and Top five Keywords.

As can be observed, there is a clear distinction in vocabulary across topics, although some overlap of words exists. This overlap likely arises because certain words commonly used in song lyrics naturally span multiple topics. Expanding the dataset with additional lyrics in future analyses could help mitigate this minor issue and further refine the distinctions between topics.

By inspection, I can also see that the probability of a song of being of a topic does make sense. Table 3 summarizes the average topic probability per artist. For instance, I can notice that most of the artists are a fair mix of the first four topics, while rappers, such as Playboi Carti, Eminem, and Tyler, The Creator are mainly addressed to Street Life topic, a fact that is perfectly coherent with reality. Finally, using another method of topic modeling, such as NMF, returns similar results. I do not report the result in this study, but it will be incorporated in the future.

Furthermore, I use DistilBERT to conduct sentiment analysis on the lyrics. DistilBERT is a transformer-based natural language processing model designed as a lightweight and faster version of BERT. It achieves comparable accuracy to BERT while being more computationally efficient, an important fact in my analysis, due to time constraints imposed by the situation this study arises from. By leveraging a pre-trained language model fine-tuned on sentiment data, DistilBERT analyzes the context and meaning of the lyrics to classify their sentiment (e.g., positive, negative, or neutral). Its ability to capture nuanced emotional tones makes it particularly effective for analyzing song lyrics. Additionally, I assess lexical richness, where a higher number of unique words used in the lyrics indicates a greater value. Figure 1 summarizes the lexical richness by artist, showing that its measure is pretty much precise, making exception for an artist, Marshmello, who is a dj and whose songs' are written and sang by different artists.

Table 3: Average Topic Probabilities Per Artist

artist	Love	Desire	Party	Reflection	Street Life
Adele	0.32	0.23	0.10	0.28	0.07
Arctic Monkeys	0.25	0.13	0.15	0.41	0.06
Bad Bunny	0.11	0.38	0.30	0.08	0.13
Beyoncé	0.20	0.11	0.15	0.19	0.36
Billie Eilish	0.19	0.14	0.23	0.38	0.06
Bruno Mars	0.18	0.16	0.14	0.24	0.28
Calvin Harris	0.28	0.16	0.14	0.22	0.21
Coldplay	0.34	0.20	0.19	0.19	0.08
Daddy Yankee	0.20	0.17	0.40	0.08	0.15
David Guetta	0.28	0.14	0.25	0.17	0.15
Doja Cat	0.04	0.06	0.08	0.10	0.73
Dua Lipa	0.15	0.13	0.28	0.38	0.06
Ed Sheeran	0.30	0.07	0.22	0.30	0.11
Elton John	0.38	0.16	0.16	0.21	0.10
Eminem	0.13	0.07	0.05	0.15	0.60
Future	0.03	0.07	0.07	0.03	0.80
Imagine Dragons	0.41	0.24	0.05	0.21	0.09
J Balvin	0.18	0.27	0.36	0.13	0.06
Karol G	0.09	0.33	0.33	0.13	0.12
Katy Perry	0.31	0.13	0.23	0.15	0.19
Khalid	0.26	0.18	0.11	0.32	0.12
Lady Gaga	0.15	0.18	0.30	0.20	0.17
Lana Del Rey	0.20	0.09	0.35	0.19	0.16
Linkin Park	0.39	0.34	0.04	0.15	0.07
Maroon 5	0.17	0.32	0.14	0.28	0.10
Marshmello	0.20	0.17	0.16	0.32	0.14
Miley Cyrus	0.15	0.17	0.21	0.30	0.17
One Direction	0.27	0.18	0.21	0.29	0.05
Pitbull	0.12	0.12	0.16	0.12	0.48
Playboi Carti	0.04	0.05	0.04	0.04	0.84
Post Malone	0.09	0.05	0.09	0.18	0.60
Queen	0.31	0.23	0.13	0.18	0.15
Rosé	0.13	0.20	0.19	0.36	0.13
SZA	0.09	0.10	0.13	0.32	0.35
Sabrina Carpenter	0.17	0.16	0.15	0.35	0.17
Sam Smith	0.24	0.19	0.17	0.34	0.05
Shakira	0.17	0.23	0.30	0.21	0.09
Shawn Mendes	0.23	0.18	0.24	0.30	0.04
Sia	0.43	0.16	0.09	0.20	0.12
Taylor Swift	0.21	0.17	0.21	0.31	0.09
Travis Scott	0.09	0.05	0.05	0.08	0.74
Tyler, the Creator	0.06	0.05	0.07	0.11	0.72

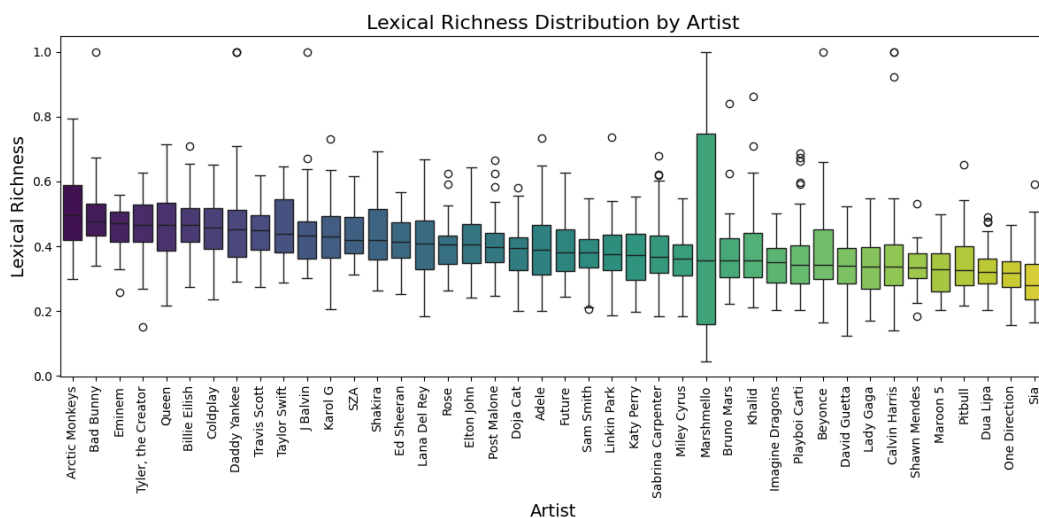


Figure 1: Lexical Richness by Artist

Finally, I undertake what is likely the most innovative aspect of this study: identifying rhetorical figures and stylistic elements that are characteristic of songs to use them as predictors.

This part of the analysis is undoubtedly the most challenging and demands further refinement, as identifying these elements is considerably more complex than previous tasks. The utility of obtaining lyrics directly from the Genius API is evident here, as it allowed me to divide the lyrics into lines (verses) to better analyze the poetic devices discussed below. I identify: alliterations, epiphoras, anaphoras, rhymes, and assonances

An *alliteration* is a rhetorical figure in which a sound is repeatedly emphasized through the recurrence of the same initial letter or sound in closely connected words. For example, an instance from recovered from the data is: *"searching and searching for someone to save my soul ooh."*

To identify alliterations, I divided the lyrics into individual lines and searched for patterns within each verse. However, this approach has a notable limitation: it may underestimate the total number of alliterations in a song, as some might occur across multiple verses rather than within a single one. I define a verse as containing an alliteration if it meets the following criteria:

- For verses containing up to 7 words, at least 4 words must begin with the same letter.
- For verses with 8 to 11 words, a minimum of 6 words must start with the same letter.
- For verses with 12 or more words, at least 8 words must share the same initial letter.

In total, I find 81 alliteration, 0.038 per song.

In addition, I identified two other rhetorical figures: anaphoras and epiphoras. An anaphora is the repetition of a word or phrase at the beginning of successive clauses, sentences, or lines, often used to create emphasis, rhythm, and cohesion within a text. In contrast, an epiphora (also called epistrophe) involves the repetition of a word or phrase at the end of successive clauses, sentences, or lines, used to emphasize a particular idea and make it more memorable. My analysis uncovered over 15,000 instances of anaphoras and more than 12,000 instances of epiphoras.

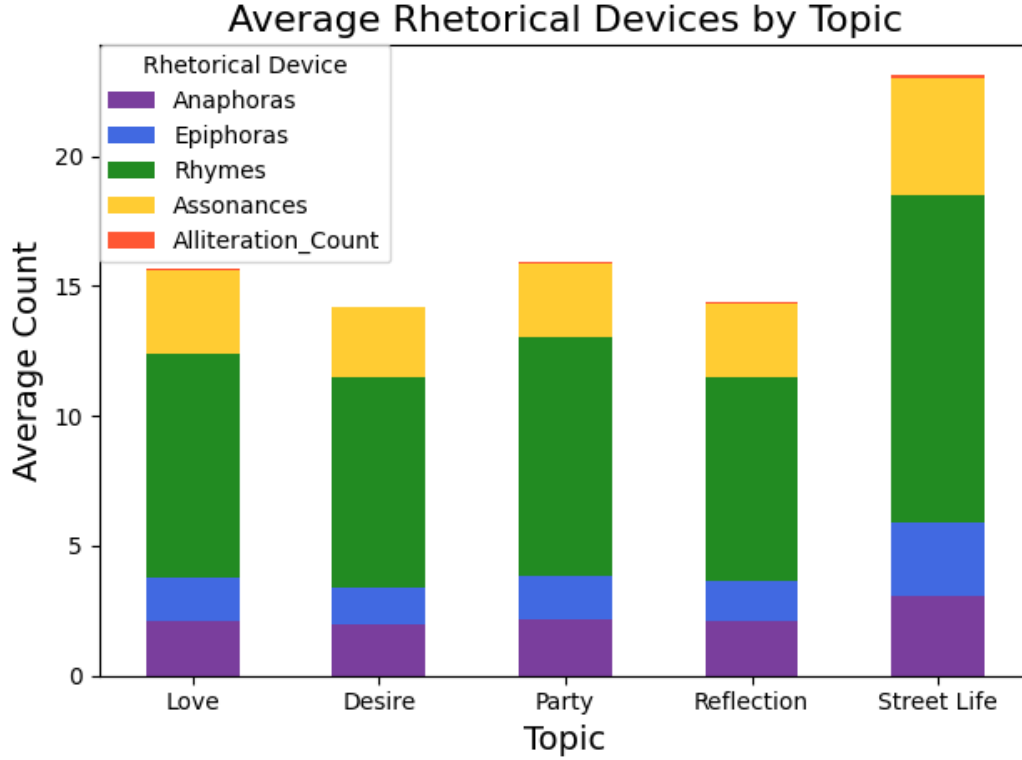


Figure 2: Average number of rhetoric figures and poetic devices in a song by topic

I also analyzed rhymes and assonances. A key distinction between a rhyme and an assonance is that a rhyme matches both vowel and consonant sounds at the end of words, while an assonance matches only vowel sounds, irrespective of the consonants that follow.

For instance, a rhyme is:

*"I was out here playing games
You was out here with a lame"*

while an assonance is:

*"Top, top, top, top of the morning like yeehaw
Thought you'd be there when I need y'all"*

Both come from Tyler, The Creator's *The Creator's Best Interest* masterpiece. Figure 2 presents the average counts of epiphoras, anaphoras, rhymes, assonances, and alliterations across topics.

This task is far from straightforward. In this study, I focus on identifying exact rhymes but plan to expand the analysis to include additional types of rhymes, such as alternate, slant, and eye rhymes, among others. I view this research as paving the way for future work to refine and broaden the range of predictors. In subsequent studies, I aim to identify a wider array of rhymes (e.g., alternate and non-exact rhymes) and explore additional rhetorical figures, such as personification and metaphors.

3.2 Kaggle Data

The second source I use is a dataset titled "Most Streamed Spotify Songs 2024", which is publicly available on Kaggle. This dataset includes 4600 songs, supposedly the most

listened songs in 2024. It stands out among other available datasets because it provides the number of streams per song, a metric that is otherwise difficult to obtain. For instance, Spotify API does not provide the number of streams. Instead, it offers a "popularity" index, which primarily reflects most recent streams. This makes it an improper measure to assess whether a song is a "hit" or not, as a song that used to be a hit 5 years ago today may be forgotten and not popular at all. Therefore as this index, easier to find, would complicate my analysis, I searched for a dataset containing the number of streams. In an ideal scenario, I would have created a custom dataset by web scraping the number of views or streams (henceforth referred to as "streams") from YouTube and Spotify, the two largest music streaming platforms. This data could then be merged with a larger and more comprehensive lyrics dataset.

The dataset also includes streams from platforms like TikTok and SoundCloud, but these sources are less complete than those for Spotify and YouTube. Consequently, I decided to retain only Spotify and YouTube data for consistency. Figure 3 illustrates the average streams per song by the release year, showing a general trend where newer songs accumulate more streams. The high values are due to the fact that I have an elite sample.

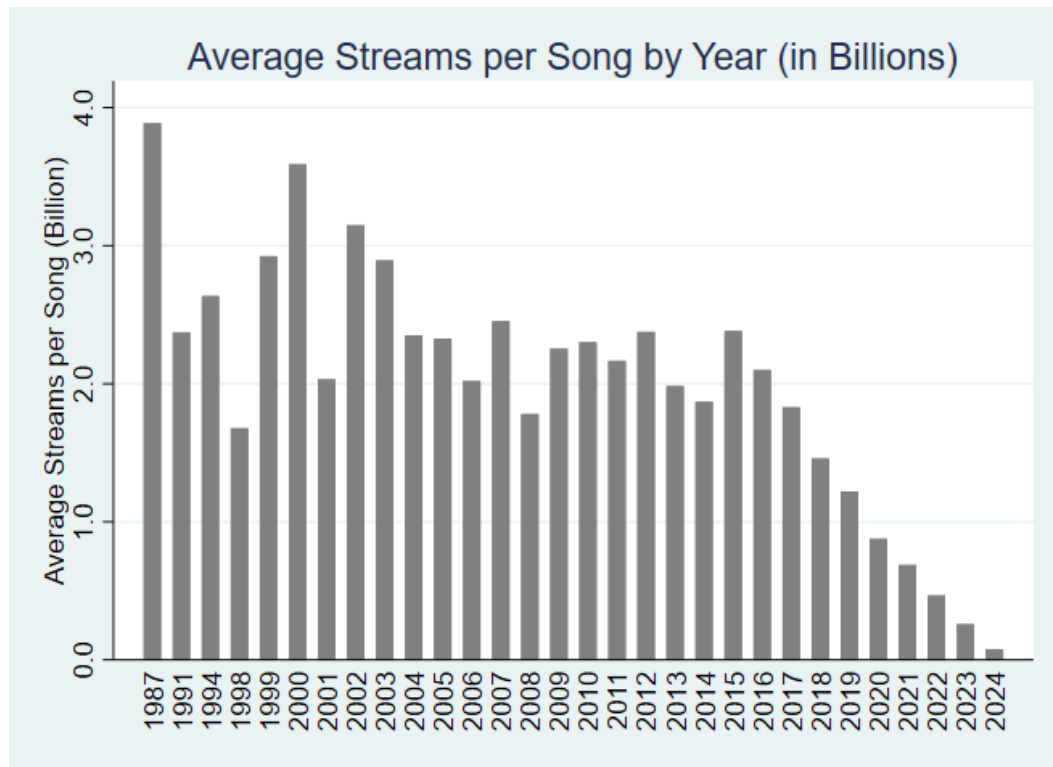


Figure 3: Average number of streams by year (in Billions)

The next section presents the results obtained after merging this dataset with the Genius lyrics data.

3.3 Merged dataset

Upon merging the two datasets, I obtained 314 observations. As emphasized earlier, this sample size is too small to yield robust or generalizable results. Expanding the number of observations in future studies will be essential. To account for the effect of time, I regress streams on time to obtain the residualized streams as the outcome variable of my study,

given that older songs are expected to have accumulated more streams.

Figure 4 presents the correlation matrix, which reveals no significant correlations between the predictors and the streams. This result is not surprising, given the limited number of features in the dataset. To improve the predictive power of the analysis, I plan to incorporate additional song metrics, such as duration, key, instrumentation, beats per minute (BPM), and more. As already pointed out, it would not be realistic to think that lyrics alone can predict the number of streams. Rather, it should be use as a feature to perfection a prediction.

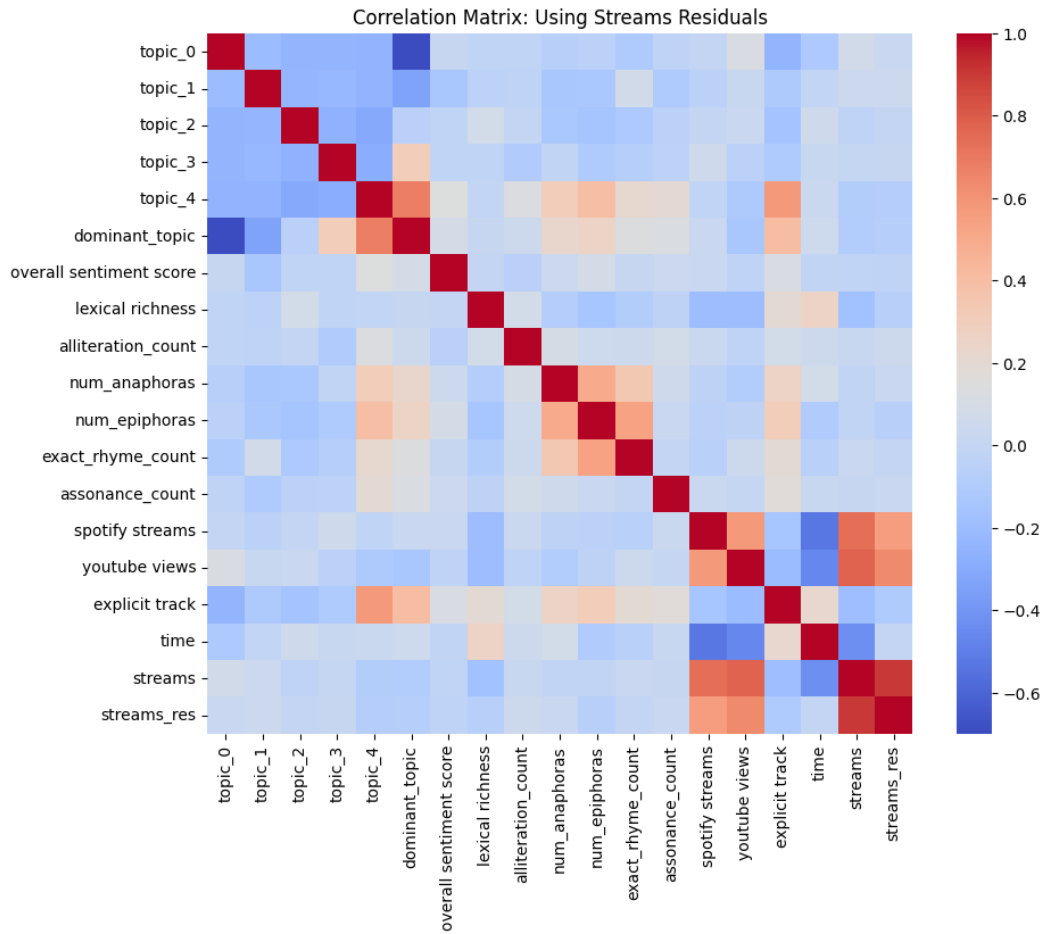


Figure 4: Correlation Matrix

Figure 5 shows the average topic distribution by year. The "Street Life" topic (Topic 4) was relatively low for several years before experiencing a resurgence starting in 2015, likely due to the popularity of trap music. However, it is important to note that these trends are neither generalizable nor representative due to the small dataset size.

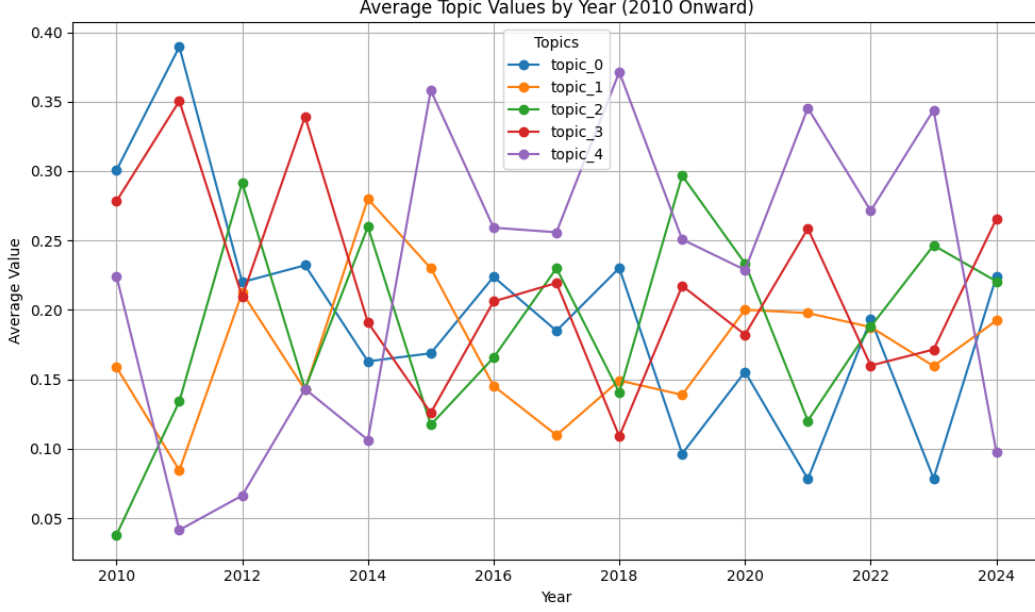


Figure 5: Topic values by years

4 Prediction

The results of this analysis indicate that the models struggled to predict the number of streams, as reflected by R-squared values near zero across all approaches. Again, this was to be expected as the it is not imaginable that the only lyrics can function as predictors for success alone, but they can probably be a further refinement. However, in this section I assess the results of my analysis. Three models were tested: Pooled OLS, Random Forest, and a Neural Network. Despite optimization efforts, none demonstrated meaningful predictive power, underscoring significant limitations in the dataset and feature set.

To establish a baseline, a Pooled OLS¹ regression was performed with training and test data pooled over time. However, the results were weak, suggesting the dataset was insufficient to explain the variance in the target variable. This outcome aligns with patterns in the correlation matrix showed in Figure 4, which revealed weak relationships between predictors and the number of streams.

Building on this, Random Forest and Neural Network models were applied, incorporating additional features such as the average past streams for each artist alongside those used in OLS. Despite attempts to standardize variables, apply log transformations, and include interaction terms, these models failed to improve performance, capturing little variance in the target variable.

The Neural Network was designed to capture non-linear relationships between features and the number of streams, using an architecture with three hidden layers (64, 32, and 16 neurons) and ReLU activation. Regularization was applied via dropout layers, and the Adam optimizer was used to minimize mean squared error (MSE) over 50 epochs. Despite efforts to optimize performance, the model struggled to generalize, as reflected by an R-squared value near zero on the test set and high root mean squared error (RMSE). The preprocessing involved imputing missing values, standardizing features, and transforming

¹All regressors are already discussed except explicit track which equals one if the song is considered explicit by Spotify and zero otherwise.

the target variable using Yeo-Johnson normalization to address skewness. The poor results were attributed to the small dataset (314 samples), insufficient variability in features, and potential noise in the predictors. Neural Networks, which typically excel with large and diverse datasets, were unable to extract meaningful patterns from the available data.

Table 4: OLS Regression Results

topic_0	0.0126 (0.0490)
topic_1	0.0247 (0.0490)
topic_2	-0.0043 (0.0460)
topic_3	0.0056 (0.0460)
topic_4	-0.0337 (0.0570)
overall sentiment score	-0.0001 (0.0580)
lexical richness	-0.0533 (0.0600)
alliteration count	0.0544 (0.0580)
anaphoras count	0.0805 (0.0670)
epiphoras count	-0.0946 (0.0780)
exact rhyme count	0.0400 (0.0690)
assonance count	0.0394 (0.0590)
explicit track	-0.0782 (0.0740)
Intercept	-5.204e-18 (0.0570)
N	314
R-squared	0.027

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A Random Forest was employed as an alternative, leveraging its ability to model non-linear interactions and feature importance. The model included 200 trees with a maximum depth

of 5 to avoid overfitting, using interaction terms, such as those between topics and lexical richness, were engineered to capture relationships between predictors. Similar to the Neural Network, predictors were scaled, and the target variable was normalized using Yeo-Johnson transformation. While the model performed reasonably on the training set, test set results showed poor generalization, with an R-squared value near zero and predictions that failed to align with actual values. The Random Forest revealed that features like average past streams and lexical richness were relatively more important, but their contributions were not strong enough to overcome the dataset’s limitations. Overall, the model underscored the challenge of predicting streams with the limited features and sample size available.

Several factors likely contributed to the poor results. The dataset was relatively small, with only 314 observations after merging, and lacked key song-level features such as tempo, key, instrumentation, and beats per minute (BPM). Additionally, limited variability in rhetorical features and sentiment scores among songs may have provided insufficient signal for predicting streams.

Table 4 presents the OLS regression results, and Table 5 compares the performance of the Random Forest and Neural Network models. These findings emphasize the need for future work to expand the dataset and incorporate richer, more diverse features to address these limitations effectively.

Table 5: Model Performance Metrics Comparison

Metric	Model	Train	Test
RMSE	Neural Network	0.8222	0.3279
	Random Forest	0.7799	0.4557
R-squared	Neural Network	0.3239	-0.4986
	Random Forest	0.3916	-1.8948

Finally, Figure 6 shows the loss curves and how they do not converge, confirming that the model is performing poorly.

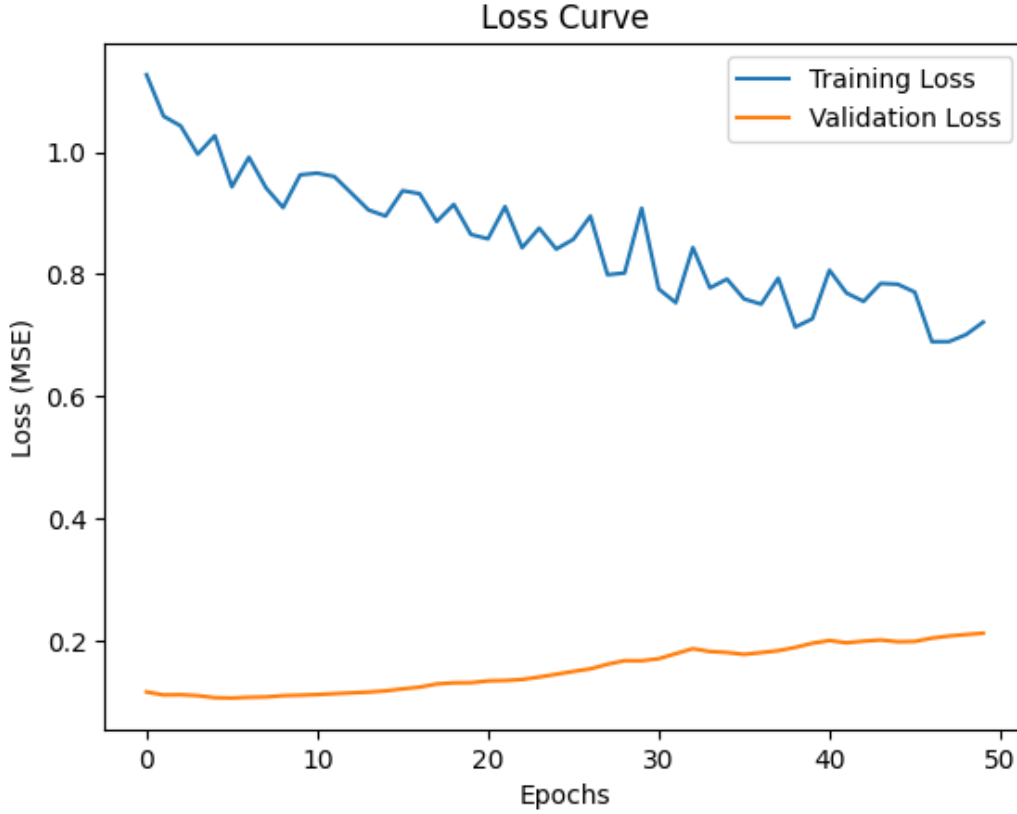


Figure 6: Training and Validation Loss curves

5 Conclusion

Predicting the success of music tracks remains a complex and multifaceted challenge, shaped by an intricate interplay of factors, including lyrical content, artist reputation, production quality, and broader cultural trends. This study explored whether lyrical attributes—such as themes, sentiments, and rhetorical devices—could serve as reliable predictors of a song’s performance. Using data from the Genius API and Spotify-related datasets, I applied advanced natural language processing techniques and predictive modeling approaches. Despite the innovative methodologies employed, my findings revealed significant limitations in the ability of lyrical content alone to predict success, as evidenced by low R-squared values and underperforming models. However, I believe that lyrics contain insightful information that must not be left unconsidered and that a deeper and perfected study may benefit from including them as part of the analysis.

The results underscore the importance of recognizing the broader context in which music is consumed. While themes and stylistic elements may influence a listener’s engagement, factors such as production quality, marketing strategies, social media influence, and even the timing of a release are equally, if not more, critical. These elements were outside the scope of this study, but they present compelling directions for future research.

By building upon this foundation, future research can develop more robust and accurate models, ultimately advancing our understanding of the drivers behind music popularity in an increasingly data-driven and competitive industry.

5.1 Future Research Directions

To address the limitations observed in this study, future research should consider the following:

- **Incorporating Acoustic Features:** While this study focused on lyrical content, integrating acoustic features—such as tempo, key, instrumentation, and beats per minute (BPM)—could provide a more holistic view of the factors driving a song’s success. Combining textual and acoustic features in a multimodal framework could enhance predictive performance.
- **Expanding Dataset Size and Diversity:** The limited dataset used in this study—both in size and scope—restricted the generalizability of the findings. Future studies should aim to compile larger and more diverse datasets, including tracks from emerging artists, niche genres, and less commercially successful songs. This would help in understanding how lyrical patterns differ across varying levels of success.
- **Exploring Contextual Variables:** External factors, such as social media engagement, marketing spend, and playlist placements, likely play a critical role in a song’s performance. Integrating these variables into predictive models could offer a more comprehensive understanding of music success.
- **Analyzing Longitudinal Trends:** Music consumption patterns evolve over time, influenced by cultural and technological shifts. Future studies could explore how the importance of lyrical themes changes across decades, genres, or geographical regions.
- **Fine-Grained Analysis of Rhetorical Devices:** This study introduced rhetorical features, such as alliteration, anaphora, and rhyme schemes, as potential predictors. Future work could refine these methods by including more nuanced features like metaphor usage, personification, and complex rhyme patterns. Additionally, examining how rhetorical complexity varies by genre could yield deeper insights.
- **Different Applications of Lyrics:** Lyrics can be used for many purposes beyond predicting a song’s success. For example, they can help study crime behavior by analyzing themes like violence in certain genres. They can also be used to detect signs of depression or emotional struggles by identifying expressions of sadness or distress.

In conclusion, while this study highlights the potential of using lyrical content to predict music success, it also emphasizes the need for more comprehensive approaches that account for the multifaceted nature of this challenge. By expanding datasets, integrating diverse features, future research can build on these foundations to deliver deeper insights into what makes a song successful. This interdisciplinary effort will not only advance the field of computational musicology but also provide valuable tools and strategies for the music industry in an increasingly data-driven era.

References

1. Aguiar, L., & Waldfogel, J. (2018). Quality Predictability and the Welfare Benefits of New Products: Evidence from the Digitization of Recorded Music. *Journal of Political Economy*, Vol. 126, No. 2, pp. 492–524.
2. Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet Allocation. *Journal of Machine Learning Research*, Vol. 3, pp. 993–1022.
3. Brynjolfsson, E., Hu, Y. J., & Smith, M. D. (2003). Consumer Surplus in the Digital Economy: Estimating the Value of Increased Product Variety at Online Booksellers. *Management Science*, Vol. 49, No. 11, pp. 1580–1596.
4. Fell, M., & Sporleder, C. (2014). Lyrics-Based Analysis and Prediction of Hit Songs. *Proceedings of the 25th International Conference on Computational Linguistics (COLING)*, pp. 620–631.
5. Malmi, E., Takala, P., Toivonen, H., Raiko, T., & Gionis, A. (2016). DopeLearning: A Computational Approach to Rap Lyrics Generation. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 195–204.
6. Rosen, S. (1981). The Economics of Superstars. *The American Economic Review*, Vol. 71, No. 5, pp. 845–858.